

Yukun Zhou

Computer Science Researcher

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Summary

Researcher in physics-based simulation, computer graphics, HCI, and game development, with experience in rod simulation, contact and friction, fluid simulation, real-time graphics, AR interaction, and machine learning systems.

Education

University of Southern California , Computer Science - Game Development • CGPA: 3.87/4.0	Los Angeles, CA, USA 2023 – 2025
City University of Hong Kong , Computer Science, Minor in Creative Media • CGPA: 3.8/4.30 • Ranked top 14 out of 144 Computer Science students. • Average 3.95/4.30 for major courses. • First Class Honors.	Hong Kong 2019 – 2023

Experience

Chinese Academy of Sciences Hong Kong Institute of Science and Innovation and Chinese University of Hong Kong , Research Assistant, Joint RA-PhD Track - Physics-Based Simulation Advised by Dr Zhongkai Zhang . • Finished main experiments and produced a paper draft on accurate rod-insertion simulation algorithms based on Projective Dynamics, Cosserat rods, Lagrangian constraint-based methods, and thickness-aware rod contact pressure and friction. • Worked on the optimization of Vertex Block Descent. • Developed and reproduced SDF soft-body contact, DiSect differentiable cutting, FEM, Projective Dynamics cutting with low-rank update methods, and lung puncturing by Warp. • Completed human brain 3D reconstruction from MRI images using neural networks.	Hong Kong July 2025 – June 2026 1 year
Georgia Institute of Technology , Summer Research Intern - Fluid Simulation with Neural and Particle Flow Maps Advised by Dr Bo Zhu . • Replicated the Neural Flow Map algorithm with Taichi. • Developed real-time 3D stable fluid simulation with OpenGL Compute Shader. • Accelerated Particle Flow Map using long-range mapping and short-range projection, achieving an 80% speed improvement.	Atlanta, GA, USA June 2024 – Sept 2024 4 months
City University of Hong Kong , Part-Time Research Assistant - HCI Research on Games and Players Advised by Dr Zhicong Lu . • Produced a paper submitted to a top-tier conference as the only first author. • Completed an HCI research project on games and ICT4D, with design implications for player well-being and future application scenarios.	Hong Kong June 2022 – Sept 2023 1 year 4 months

City University of Hong Kong, Part-Time Research Assistant - Hand-Scratching Gesture Recognition Android Application for AR Glasses
Advised by Dr Kening Zhu.

Hong Kong
July 2022 – May 2023
11 months

- Used CNN models to analyze scratching sounds on AR glasses legs in real time.
- Developed an Android application for recording, real-time recognition, and AR glasses control.

University of Southern California, Grader, Physics-Based Simulation and Animation
Grader under [Dr. Jernej Barbic](#) for [CSCI 520: Computer Animation and Simulation](#).

Los Angeles, CA, USA
Jan 2025 – May 2025
5 months

Siemens Mobility, Data Science Intern

Hong Kong
June 2021 – Mar 2022
10 months

- Designed and developed map matching, route matching, traffic flow prediction algorithms, and time-series models including bi-LSTM, encoder-decoder LSTM, and ARIMA.
- Manipulated databases and processed data with Python to support database building and maintenance.

Publications

About the Optimization of Vertex Block Descent

2026

Anonymous Author, Yukun Zhou, Other Authors
(Manuscript submitted)

Thickness-Aware Rod Insertion Simulation with Rotational Friction

2026

Yukun Zhou, Other Authors
(Manuscript in preparation)

Awards

Entrance Full Tuition Scholarship

City University of Hong Kong
2019-2023

Dean's List Awards for Outstanding Academic Performance

City University of Hong Kong
2019-2023

Projects

Physics-Based Simulations

Implemented mass-spring systems, keyframe interpolation with quaternion and Euler approaches, inverse kinematics with pseudo-inverse and Tikhonov regularization methods, and related simulation components in C++.

Unity Game Development

Led, designed, and developed Unity 3D game projects including Lucidmare, PL-23, and Donuts!, with Donuts! published on Steam.

3D Human Rendering

Built a 3D graphics course project with URP, hair anisotropy, and skin separable subsurface scattering under Unity, including GLSL deployment of skin separable subsurface scattering.

NLP for Finance

Advised by Dr Linqi Song at City University of Hong Kong. Developed a BERT model in PyTorch to analyze sentiment across large-scale financial comments.

Skills

Simulation and Graphics: Projective Dynamics, FEM, Cosserat rods, SDF contact, differentiable cutting, OpenGL Compute Shader, Taichi, Unity, GLSL

Machine Learning: CNN, BERT, PyTorch, neural 3D reconstruction, time-series modelling, bi-LSTM, encoder-decoder LSTM, ARIMA

Programming: Python, C, Java, C++, Processing, JavaScript, HTML, CSS, SQL

Other Technical Skills: AI programming, drone programming, digital art and installation, game design and development

Languages

Mandarin

Native

English

Fluent

Cantonese

Basic